William Miller

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EDUCATION

Tufts University Medford, MA
Bachelor of Science in Mechanical Engineering *Expected: May 2028*
GPA: 3.98

Relevant Courses: Mechanics I & II, Thermal Fluid Systems I & II, Mechanical Design I & II, Electronics and Controls, Computer-Aided Product Design, Materials and Manufacturing, Differential Equations, Multivariable Calculus, Numerical Methods

ENGINEERING EXPERIENCE

Tufts Solar Vehicle Project Medford, MA
Dynamics Lead *Apr 2026 – Present*
Team Member *Sept 2025 – Apr 2026*

- Designed a roll cage in SolidWorks and developed a PVC prototype with 3D-printed elbow joints to validate geometry, then redesigned for manufacturability after identifying bending limitations with available tooling
- Performed FEA on the front hoop across 5 required load cases (combined 5g/4g/1.5g loading, 30° and 60° angled loading, lateral and rearward horizontal loading), confirming stresses remained within yield strength per competition safety regulations

Chiesa Research Group (Superconductivity Lab) Medford, MA
Undergraduate Research Assistant *June 2026 – Present*

- Co-design and machined custom Instron grips (CAD, sizing for strength and fit, lathe work) to enable round-sample mechanical testing for an external industry client
- Diagnosed systematic measurement error in a video extensometer caused by cylindrical sample geometry and developed a modified camera-calibration procedure that reduced an ~5mm gauge-length error to within 1 mm

WORK EXPERIENCE

Tufts Department of Physics Medford, MA
Learning Assistant *Sept 2025 – Dec 2025*

- Selected by course instructor to guide students through complex physics problems during lectures and office hours, fostering discussion through targeted questioning and encouraging multiple solution approaches
- Attended weekly instructor meetings to provide feedback on course dynamics and student progress

Associates in Ophthalmology Livingston, NJ
Ophthalmic Technician / Administrative Assistant *June 2020 – Aug 2023*

- Conducted preliminary eye exams using medical diagnostic equipment in a regulated clinical environment, emphasizing precision, documentation, and patient safety
- Digitized years of patient records into a new electronic health records system, improving office efficiency and data accessibility

PROJECTS

Motor-Controlled Clock In Progress

- Designed a fully custom clock mechanism in SolidWorks, including a gear train sized for stepper motor actuation, and 3D printed the majority of components
- Developing stepper motor timekeeping logic and finalizing motor mount geometry pending motor measurements for full assembly

Assistive Paint Dispenser Apr 2026 – May 2026

- Designed an assistive paint dispenser in SolidWorks for a user with cerebral palsy, accommodating two paint tube sizes with an ergonomic 3D-printed handle and die, and aluminum frame manufactured via waterjet cutting
- Prototyped iteratively using laser-cut acrylic to validate geometry before final fabrication, and produced custom paint trays via vacuum forming to complete the accessible workspace solution

SKILLS

CAD & Simulation: SolidWorks (CSWA certification), Onshape, FEA
Programming & Analysis: MATLAB, C++, Excel
Manufacturing & Prototyping: 3D printing, laser cutter, water jet cutter, manual mill, manual lathe (including threading)